

Lab04

AUTHOR

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```
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

```
filter, lag
```

Os seguintes objetos são mascarados por 'package:base':

```
intersect, setdiff, setequal, union
```

```
library(readr)  
library(lubridate)
```

Anexando pacote: 'lubridate'

Os seguintes objetos são mascarados por 'package:base':

```
date, intersect, setdiff, union
```

1) Carregue os pacotes readxl e tidyverse

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —  
✓ forcats 1.0.1      ✓ stringr 1.6.0  
✓ ggplot2 4.0.3      ✓ tibble  3.3.1  
✓ purrr  1.2.2      ✓ tidyr   1.3.2  
— Conflicts — tidyverse_conflicts() —  
✖ dplyr::filter() masks stats::filter()  
✖ dplyr::lag()     masks stats::lag()  
! Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)
```

2) Crie uma variável chamada fname que contenha o caminho para o arquivo WDIEXCEL.xlsx, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando file.exists

```
fname <- "WDIEXCEL.xlsx"  
file.exists(fname)
```

[1] TRUE

3) Leia a planilha "Country" e armazene no objeto countries.

```
countries <- read_excel(fname, sheet = "Country")
head(countries)
```

```
# A tibble: 6 × 30
  `Country Code` `Short Name`      `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>          <chr>        <chr>      <chr>
1 ABW           Aruba           Aruba        Aruba      AW
2 AFE           Africa Eastern and Sou... Africa East... Africa Eas... ZH
3 AFG           Afghanistan      Afghanistan  Islamic St... AF
4 AFW           Africa Western and Cen... Africa West... Africa Wes... ZI
5 AGO           Angola           Angola       People's R... AO
6 ALB           Albania          Albania      Republic o... AL
# i 25 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   `Region` <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

4) Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha).

```
countries1 <- countries %>% filter(`Short Name` %in% c("Brazil", "Argentina", "Japan", "France"))
head(countries1)
```

```
# A tibble: 5 × 30
  `Country Code` `Short Name`      `Table Name` `Long Name`      `2-alpha code`
  <chr>          <chr>          <chr>        <chr>          <chr>
1 ARG           Argentina      Argentina    Argentine Republic AR
2 BRA           Brazil        Brazil       Federative Republic o... BR
3 FRA           France        France       French Republic     FR
4 ITA           Italy         Italy        Italian Republic     IT
5 JPN           Japan         Japan        Japan                JP
# i 25 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   `Region` <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

5) Leia a planilha "Data" e armazene no objeto WDI.

```
WDI <- read_excel(fname, sheet = "Data")
head(WDI)
```

```
# A tibble: 6 × 66
  `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
```

```

<chr>      <chr>      <chr>      <chr>      <dbl> <dbl>
1 Africa Eastern... AFE      Access to clean... EG.CFT.ACCS.ZS      NA      NA
2 Africa Eastern... AFE      Access to clean... EG.CFT.ACCS.RU.... NA      NA
3 Africa Eastern... AFE      Access to clean... EG.CFT.ACCS.UR.... NA      NA
4 Africa Eastern... AFE      Access to elect... EG.ELC.ACCS.ZS      NA      NA
5 Africa Eastern... AFE      Access to elect... EG.ELC.ACCS.RU.... NA      NA
6 Africa Eastern... AFE      Access to elect... EG.ELC.ACCS.UR.... NA      NA
# i 60 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
#   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
#   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
#   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
#   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>,
#   `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, `1995` <dbl>, ...

```

6)Selecione apenas as colunas: - Country Code - Indicator Code - E todos os anos de 1960-2018

```
WDI1 <- WDI %>% select(`Country Code`, `Indicator Code`, `1960`:`2018`)
```

7)Confirme se os dados wm WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

```
head(WDI1)
```

```

# A tibble: 6 × 61
  `Country Code` `Indicator Code` `1960` `1961` `1962` `1963` `1964` `1965`
  <chr>          <chr>          <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 AFE           EG.CFT.ACCS.ZS      NA     NA     NA     NA     NA     NA
2 AFE           EG.CFT.ACCS.RU.ZS   NA     NA     NA     NA     NA     NA
3 AFE           EG.CFT.ACCS.UR.ZS   NA     NA     NA     NA     NA     NA
4 AFE           EG.ELC.ACCS.ZS      NA     NA     NA     NA     NA     NA
5 AFE           EG.ELC.ACCS.RU.ZS   NA     NA     NA     NA     NA     NA
6 AFE           EG.ELC.ACCS.UR.ZS   NA     NA     NA     NA     NA     NA
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
#   `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
#   `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
#   `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>,
#   `1995` <dbl>, `1996` <dbl>, `1997` <dbl>, `1998` <dbl>, `1999` <dbl>, ...

```

```
WDI1 <- WDI1 %>% pivot_longer(cols = `1960`:`2018`, names_to = "YEAR", values_to = "VALUE")
head(WDI1)
```

```

# A tibble: 6 × 4
  `Country Code` `Indicator Code` YEAR  VALUE
  <chr>          <chr>          <chr> <dbl>
1 AFE           EG.CFT.ACCS.ZS   1960    NA
2 AFE           EG.CFT.ACCS.ZS   1961    NA
3 AFE           EG.CFT.ACCS.ZS   1962    NA
4 AFE           EG.CFT.ACCS.ZS   1963    NA

```

5 AFE	EG.CFT.ACCS.ZS	1964	NA
6 AFE	EG.CFT.ACCS.ZS	1965	NA

inicialmente os dados não estavam em formato tidy

8) Confirme se a coluna referente ao ano está em formato inteiro.

```
class(WDI1$YEAR)
```

```
[1] "character"
```

```
WDI1$YEAR <- as.integer(WDI1$YEAR)
```

9) Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas: - Country Code - Short Name - YEAR - SP.DYN.CBRT.IN

```
WDI1 <- WDI1 %>% filter(`Country Code` %in% countries$`Country Code`,
  `Indicator Code` == "SP.DYN.CBRT.IN") %>%
  left_join(countries %>% select(`Country Code`, `Short Name`), by = "Country Code") %>% select
```

10) Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

```
WDI1 <- WDI1 %>% arrange(`Short Name`, YEAR)
```

11) Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```
library(ggplot2)
WDI1 <- WDI1 %>% rename(`SP.DYN.CBRT.IN` = VALUE)
ggplot(WDI1, aes(x = YEAR, y = `SP.DYN.CBRT.IN`, color = `Short Name`, group = `Short Name`))
  geom_line() +
  labs(x = "Ano", y = "Taxa bruta de natalidade", color = "País") +
  theme_minimal()
```

Warning: Removed 579 rows containing missing values or values outside the scale range (`geom\_line()`).

	Ireland	Latvia	-
	Isle of Man	Least developed countries: UN classification	-
	Israel	Lebanon	-
	Italy	Lesotho	-
oor countries (HIPC)	Jamaica	Liberia	-
	Japan	Libya	-
	Jordan	Liechtenstein	-
China	Kazakhstan	Lithuania	-
	Kenya	Low & middle income	-
	Kiribati	Low income	-
	Korea	Lower middle income	-
	Kosovo	Luxembourg	-
	Kuwait	Macao SAR, China	-
	Kyrgyz Republic	Madagascar	-
	Lao PDR	Malawi	-
	Late-demographic dividend	Malaysia	-
	Latin America & Caribbean	Maldives	-
	Latin America & Caribbean (excluding high income)	Mali	-
	Latin America & Caribbean (IDA & IBRD)	Malta	-

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-09 19:57:42 -03"
```